

Linux-assignments-solutions

Tuesday, February 11, 2025 4:15 PM

Part A: Practical Questions

Perform the following tasks on your HEC2 instance (or your local lab environment) and record your results. Provide screenshots or command outputs along with a brief explanation of your findings.

1. Checking Connectivity with ping

- **Task:** Use the ping command to check the connectivity to google.com.
- **Questions:**
 - What is the average round-trip time (RTT) reported by the command?
 - Did you observe any packet loss?
- **Answer:**

```
ubuntu@ip-172-31-89-176:~$ ping -c 5 google.com
PING google.com (172.253.63.113) 56(84) bytes of data.
64 bytes from bi-in-f113.1e100.net (172.253.63.113): icmp_seq=1 ttl=104 time=1.35 ms
64 bytes from bi-in-f113.1e100.net (172.253.63.113): icmp_seq=2 ttl=104 time=1.49 ms
64 bytes from bi-in-f113.1e100.net (172.253.63.113): icmp_seq=3 ttl=104 time=1.29 ms
64 bytes from bi-in-f113.1e100.net (172.253.63.113): icmp_seq=4 ttl=104 time=1.50 ms
64 bytes from bi-in-f113.1e100.net (172.253.63.113): icmp_seq=5 ttl=104 time=1.20 ms

--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.204/1.368/1.503/0.114 ms
ubuntu@ip-172-31-89-176:~$
```

RTT is 1.368ms.

Packet loss- no packet loss was observed, indicating stable connectivity.

2. Tracing the Route with traceroute/tracert

- **Task:** Use traceroute (Linux/macOS) or tracert (Windows) to map the path your packets take to reach google.com.
- **Questions:**
 - Identify and list the IP addresses of the first three hops.
 - Comment on any noticeable delays in any of the hops.
- **Answer:**
First three hops -
1) 240.3.180.77
2) 99.82.181.22
3) 72.14.203.158
No noticeable delay at the second hop, possibly due to ISP routing policies.

```
ubuntu@ip-172-31-89-176:~$ traceroute google.com
traceroute to google.com (172.253.63.113), 30 hops max, 60 byte packets
 1 240.3.180.77 (240.3.180.77) 2.231 ms 240.3.180.78 (240.3.180.78) 2.198 ms 240.3.180.77 (240.3.180.77) 2.204 ms
 2 99.82.181.22 (99.82.181.22) 2.139 ms 2.123 ms *
 3 72.14.203.158 (72.14.203.158) 2.075 ms 99.82.181.23 (99.82.181.23) 1.957 ms 1.939 ms
 4 * 216.239.62.111 (216.239.62.111) 1.894 ms *
 5 192.178.248.38 (192.178.248.38) 2.128 ms * 142.251.69.208 (142.251.69.208) 3.344 ms
 6 142.251.49.71 (142.251.49.71) 1.810 ms 142.251.49.194 (142.251.49.194) 2.175 ms 192.178.248.40 (192.178.248.40) 12.361 ms
 7 142.251.226.101 (142.251.226.101) 3.591 ms 142.251.226.93 (142.251.226.93) 2.929 ms 216.239.50.97 (216.239.50.97) 2.108 ms
 8 142.251.236.105 (142.251.236.105) 2.094 ms 142.250.209.248 (142.250.209.248) 2.881 ms 216.239.46.31 (216.239.46.31) 2.060 ms
 9 172.253.72.1 (172.253.72.1) 2.850 ms 172.253.72.71 (172.253.72.71) 3.220 ms 108.170.236.85 (108.170.236.85) 2.000 ms
10 * 172.253.72.29 (172.253.72.29) 2.543 ms 172.253.72.3 (172.253.72.3) 1.945 ms
11 * * *
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * bi-in-f113.1e100.net (172.253.63.113) 1.862 ms *
ubuntu@ip-172-31-89-176:~$
```

3. Monitoring Open Ports with netstat

- **Task:** Use netstat to display all active connections and listening ports on your instance.
- **Questions:**
 - Identify which ports are open and list the services associated with common ports like SSH (22) and HTTP (80).
 - Explain what the output tells you about your network's current activity.
- **Answer**

```

ubuntu@ip-172-31-89-176:~$ sudo netstat -tulpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:6010         0.0.0.0:*               LISTEN      21022/sshd: ubuntu@
tcp        0      0 0.0.0.0:80            0.0.0.0:*               LISTEN      4632/nginx: master
tcp        0      0 127.0.0.54:53         0.0.0.0:*               LISTEN      4620/systemd-resolv
tcp        0      0 127.0.0.53:53         0.0.0.0:*               LISTEN      4620/systemd-resolv
tcp6       0      0 :::22                 :::*                    LISTEN      1/systemd
tcp6       0      0 :::80                 :::*                    LISTEN      4632/nginx: master
tcp6       0      0 :::1:6010             :::*                    LISTEN      21022/sshd: ubuntu@
udp        0      0 127.0.0.1:323         0.0.0.0:*               4651/chronyd
udp        0      0 127.0.0.54:53         0.0.0.0:*               4620/systemd-resolv
udp        0      0 127.0.0.53:53         0.0.0.0:*               4620/systemd-resolv
udp        0      0 172.31.89.176:68      0.0.0.0:*               4664/systemd-networ
udp6       0      0 :::1:323              :::*                    4651/chronyd
ubuntu@ip-172-31-89-176:~$

```

Open Ports:

Port 22 (SSH) - service sshd
 Port 80 (HTTP) - service nginx
 Port 53 (DNS) - systemd-resolved
 Port 323 (NTP) - chronyd
 Port 68 (DHCP Client) - systemd-networ

4. Fetching HTTP Headers with curl

- **Task:** Use the curl command to retrieve the HTTP headers from <https://www.google.com>.
- **Questions:**
 - What are the key HTTP response headers you observe?
 - What do these headers indicate about the website's server or security configuration?

```

ubuntu@ip-172-31-89-176:~$ curl -I https://www.google.com
HTTP/2 200
content-type: text/html; charset=ISO-8859-1
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-ROLOEs1dm_F-KSfv1FmtWg' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https://report-uri https://csp.withgoogle.com/csp/gws/other-hp
accept-ch: Sec-CH-Prefers-Color-Scheme
p3p: CP="This is not a P3P policy! See g.co/p3phelp for more info."
date: Wed, 12 Feb 2025 01:23:41 GMT
server: gws
x-xss-protection: 0
x-frame-options: SAMEORIGIN
expires: Wed, 12 Feb 2025 01:23:41 GMT
cache-control: private
set-cookie: AEC=AVcja2exY8c-VbeFpWZC45v0oBk6NXJqfNh97r7F5FJ5hH4z80zHfTwMMw; expires=Mon, 11-Aug-2025 01:23:41 GMT; path=/; domain=.google.com; Secure; HttpOnly; SameSite=lax
set-cookie: NID=521=h9z703QLKxBAEIAS1ft1hZVI9ozc6wbGkVN79RTwcJj-KzntfisaZyWfneuQKbYP7UP67nPW-mGT0bkoELpcrF0k7p3SCGoEVi6enIwHAEvfND1d0WLjsXmkSU4xf7Z1A7yD_bvS5LDAUujZA87EufSsklrSh9Ylfi02XMtDfWPDsBhI04r08ZjXUvBb4nCwJ8f3wiUH3dSrZqY; expires=Thu, 14-Aug-2025 01:23:41 GMT; path=/; domain=.google.com; HttpOnly
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000
ubuntu@ip-172-31-89-176:~$

```

- **Key HTTP Response Headers:** HTTP/2 200, Content-Type: text/html, Cache-Control: private, Strict-Transport-Security.
- **Indications:** These headers indicate HTTPS security enforcement, caching behavior, and content type served by Google's server.

5. Domain Name Resolution with dig/nslookup

- **Task:** Use dig or nslookup to resolve the domain example.com to its IP address.
- **Questions:**
 - What IP address is returned for example.com?
 - Briefly explain what additional information (if any) the tool provides about the domain.
- **Answer:**
 IP address: 127.0.0.53
 Provides authoritative name servers, TTL values and DNS query timing.

```

ubuntu@ip-172-31-89-176:~$ dig example.com

;<<> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<> example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 47557
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                 35      IN      A      96.7.128.198
example.com.                 35      IN      A      23.192.228.80
example.com.                 35      IN      A      23.192.228.84
example.com.                 35      IN      A      23.215.0.136
example.com.                 35      IN      A      23.215.0.138
example.com.                 35      IN      A      96.7.128.175

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Feb 12 01:32:20 UTC 2025
;; MSG SIZE rcvd: 136

```

Theory Questions

6. TCP vs. UDP

- **Question:** Explain the differences between TCP and UDP. In what scenarios might a DevOps engineer choose UDP over TCP, and why?
Answer: TCP (Transmission Control Protocol) is connection-oriented, ensuring reliable data transfer with error checking and retransmission. It's used for applications requiring data integrity, like HTTP, SSH, and FTP.
UDP (User Datagram Protocol) is connectionless, faster but less reliable, as it doesn't guarantee delivery or order. Suitable for real-time applications like video streaming, VoIP, and online gaming.
Use Case: A DevOps engineer might choose UDP for DNS queries or streaming services due to its low latency.

7. Firewall Fundamentals

- **Question:** Describe how a firewall works in a network environment. Compare and contrast a packet filtering firewall with a stateful firewall, highlighting their pros and cons in managing network security.
Answer: A firewall controls incoming and outgoing network traffic based on security rules.
 - **Packet Filtering Firewall:** Inspects packets individually without context. Fast but less secure.
 - **Stateful Firewall:** Tracks active connections, providing better security by understanding traffic state. **Pros/Cons:** Stateful firewalls offer superior security but can be resource-intensive, while packet filters are faster but less thorough.

8. Understanding Ports

- **Question:** What are ports in the context of computer networking? Use a real-world analogy (for example, comparing ports to apartment numbers) to explain how ports facilitate communication between applications.
Answer: Ports are logical endpoints for network communication, like apartment numbers in a building. The IP address is the building's address, while ports direct traffic to the correct application (e.g., port 80 for HTTP, port 22 for SSH).

9. IPv4 vs. IPv6

- **Question:** Compare IPv4 and IPv6 addressing. What are the main differences in their structures and capacities? Why is the transition to IPv6 important for modern networks?
Answer: IPv4 uses 32-bit addresses (e.g., 192.168.1.1), supporting ~4.3 billion devices. IPv6 uses 128-bit addresses (e.g., 2001:0db8:85a3::8a2e:0370:7334), supporting an almost unlimited number of devices.
Importance of IPv6: Addresses exhaustion in IPv4, supports better routing, security, and is essential for the growth of IoT.

10. IP Address vs. MAC Address

- **Question:** Explain the roles of IP addresses and MAC addresses in network communication. How do they work together to ensure that data is correctly transmitted both within a local network and over the internet?
Answer: An IP address identifies devices on a network globally, like a postal address. A MAC address is a unique identifier for network interfaces, like a device's fingerprint.
Working Together: In local networks, devices use MAC addresses for communication. For internet communication, IP addresses route data globally, while MAC ensures delivery within local segments.